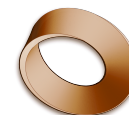
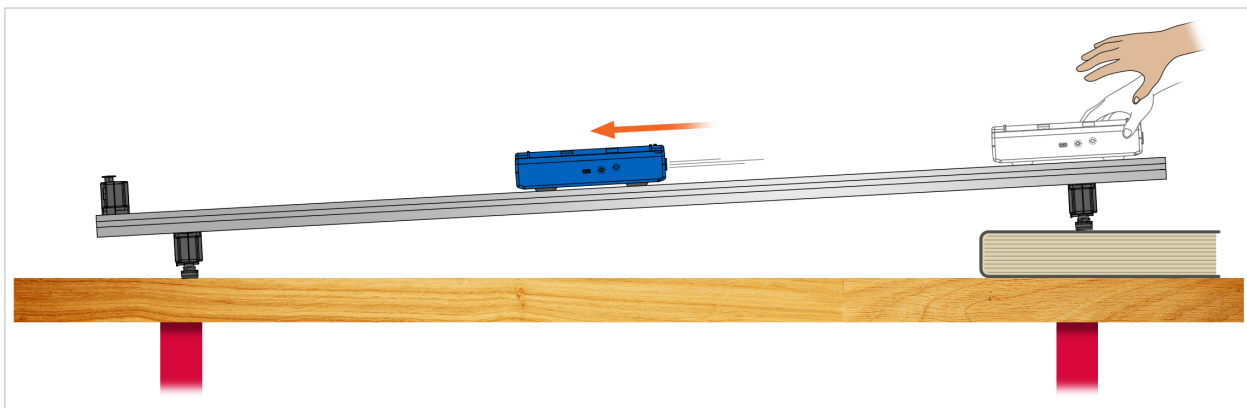


Name: _____
Class: _____
Date: _____



Mr. Rawson • GCSE Physics

Kinematics L3: Acceleration Down a Ramp



Release the cart from the top and let it run down. Equipment illustration courtesy of PASCO scientific.

Set up

1. Keep the same cart and the same laptop graphs as Experiment 1 — position and velocity, both against time.
2. Put **one book** under one end of the track. A small slope is enough. A steep one just means the cart is gone before the laptop has drawn anything.
3. Fit the **end stop at the bottom** of the track. The cart must have something to hit that is not the floor.
4. Turn the cart so the **+X arrow points down the slope**. Rolling down is now the positive direction.
5. Practice the release once, with no recording: hold the cart at the top, **let go without pushing**, and let it run into the end stop.

Let go — do not push. The whole experiment is about what gravity does to the cart on its own. A push at the start puts a jump in the graph that is nothing to do with the ramp, and you will be able to see it.

Name: _____

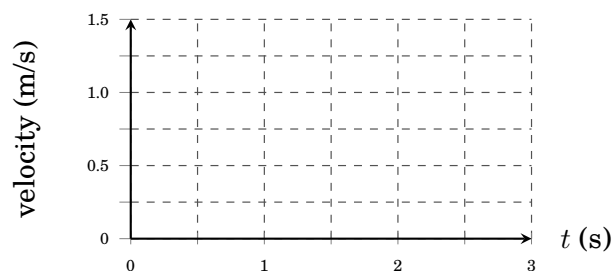
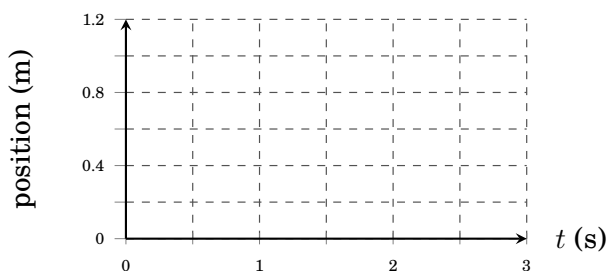
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The runs

Tick your prediction first, then run it, then sketch what the screen showed. Leave your prediction as it is.

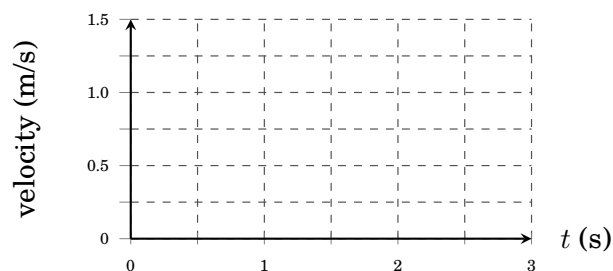
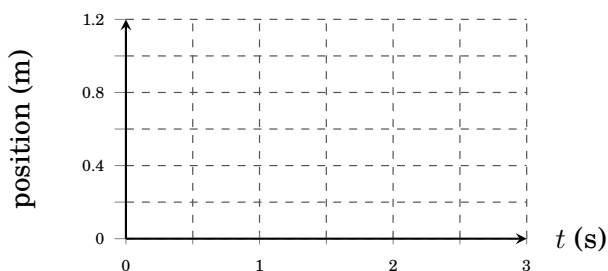
Run 1. One book. Hold at the top, start recording, let go.

Predict the *velocity* line: ☐ on the zero line ☐ zero slope, above ☐ zero slope, below ☐ tilting up ☐ tilting down



Run 2. Two books — a steeper ramp. Same release.

Predict the *velocity* line: ☐ on the zero line ☐ zero slope, above ☐ zero slope, below ☐ tilting up ☐ tilting down



Now compare the two velocity lines you have just drawn.

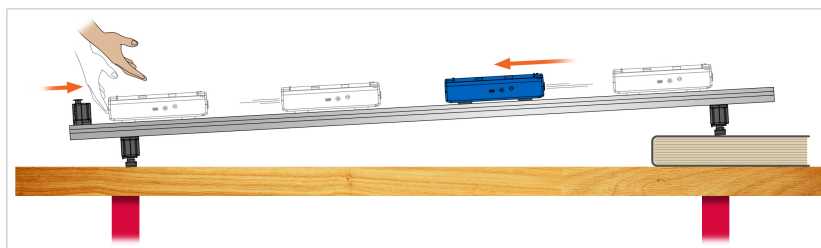
1) **Describe** what *both* velocity lines have in common, and how they differ.

2) **Explain** what the position–time line was doing in run 1, and why it was not straight.

Name: _____

Date: _____

Run 3 — up the ramp, and back down



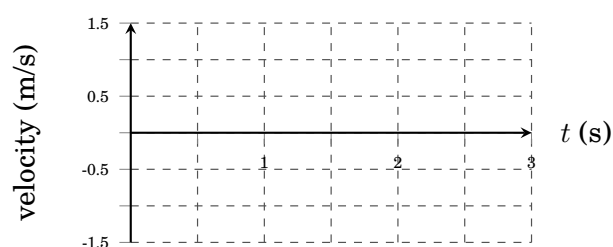
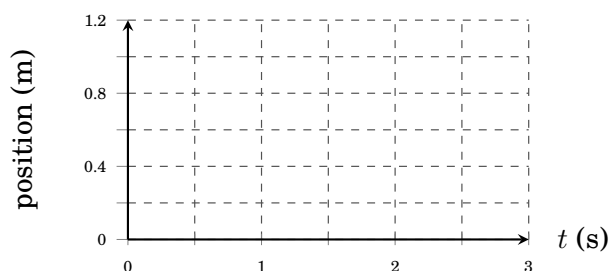
Push it up the slope and let it come back. Illustration: PASCO scientific.

Put the ramp back to **one book**. Start recording, then push the cart **up** the slope from the bottom and let it roll all the way back down. Stop recording before it reaches the end stop. **Catch it.**

For this run only, turn the cart so +X points up the slope — up is now positive, so your position graph climbs from zero and comes back to it.

Run 3. Pushed up the slope, slows, stops, comes back down.

Predict the velocity line: ☐ on the zero line ☐ zero slope, above ☐ zero slope, below ☐ tilting up ☐ tilting down



1) **Identify** the moment the cart was momentarily still. What was the velocity line doing at exactly that moment?

2) **Explain** why the velocity line crosses the zero line in this run, when it never did in Experiment 1.

3) **Describe** the shape of the velocity line over the whole run, in one sentence.

Finish the sentence. On a velocity–time graph, a line with a **slope of zero** means the speed is _____, and a line that **tilts** means the speed is _____.
The **steeper** the tilt, the _____.